Operational Runbook — Edge Inference Pipeline

Audience: distribution-center operations staff (no raw kubectl). Platform-team-only steps are marked **[PLATFORM]**.

SLA / Target Reference (fill before use)

Metric	Target
Rollout success rate (fleet-wide)	TODO %
Sync window after reconnection	TODO minutes for buffer to fully drain
MTTR — single-site incident	TODO minutes
MTTR — fleet-wide incident	TODO hours
Max concurrent site incidents before escalation	TODO
Inference latency p95 threshold	< 200 ms

1. Health Check (run first, every time)

- [ ] Open telemetry dashboard → site status panel.
- [ ] Confirm inference latency p95 < 200 ms. If above, go to §3.
- [ ] Confirm buffer utilization < 80% of 5 GB reserved. If above, go to §3.
- [ ] Confirm last successful sync timestamp < TODO minutes ago.

2. Trigger a Rollout

- [ ] Confirm change is merged to the GitOps release branch (Flux v2 AKS extension reconciles automatically — do not push manually to the cluster).
- [ ] Watch rollout status on the dashboard (Canary: Flagger analysis panel / Blue-Green: traffic-weight panel).
- [ ] Verify success criteria: TODO (e.g., error rate < X% over Y minutes).

3. Troubleshoot Common Failures

Symptom	Likely cause	Operator action
Latency p95 above threshold	Device resource contention	Restart inference module via operator dashboard action (not kubectl)
Buffer near capacity	Extended outage beyond 72h	Escalate to platform team **[PLATFORM]**

Symptom	Likely cause	Operator action
Sync not resuming after reconnect	Idempotency key collision / route misconfig	Escalate to platform team **[PLATFORM]**
Rollout stuck in Canary analysis	Metric threshold breach	Automatic rollback triggers; confirm via dashboard, no operator action needed

4. Rollback

- [ ] Identify last known-good release tag in GitOps release history.
- [ ] Trigger rollback via git revert on the release branch (Flux reconciles automatically) — operator-facing action, no kubectl.
- [ ] Confirm health check (§1) passes post-rollback.
- [ ] If rollback does not resolve within TODO minutes, escalate **[PLATFORM]**.

5. Escalation Path

1. Operator → Site Lead (immediate, if health check fails twice in a row)
2. Site Lead → Platform Team on-call (if §3/§4 doesn't resolve within MTTR target)
3. Platform Team → raw kubectl / cluster-level diagnostics (platform-only, never operator-facing)

6. [PLATFORM] Diagnostic Commands (not for operator use)

```bash
# TODO — platform-team-only kubectl/az diagnostics
kubectl get pods -n edge-inference
az iot hub device-identity connection-string show --device-id TODO
```